

STAT 412 HW07 — Question 6

Problem

The average height of females in the freshman class of a certain college has historically been 162.8 centimeters with a standard deviation of 6.8 centimeters. Is there reason to believe that there has been a change in the average height if a random sample of 48 females in the present freshman class has an average height of 165.8 centimeters? Use a P-value in your conclusion. Assume the standard deviation remains the same.

Solution

The word “changed” makes this a two-tailed test:

$$H_0 : \mu = 162.8, \quad H_1 : \mu \neq 162.8.$$

Since the population standard deviation is assumed known, use a z statistic:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{165.8 - 162.8}{6.8/\sqrt{48}} \approx 3.0566.$$

Rounded to two decimals, the test statistic is

$$z = 3.06.$$

The P-value is two-tailed:

$$P\text{-value} = 2P(Z > 3.0566) \approx 0.0022389.$$

The platform prompt says to round the P-value to three decimal places, so enter

$$0.002.$$

The value 0.0022 is the four-decimal version, but it is not rounded to the requested three decimals.

Because H_0 is rejected whenever α is at least the P-value, H_0 would be rejected for significance levels greater than the P-value.

Final Answer

$H_0 : \mu = 162.8$; $H_1 : \mu \neq 162.8$. Test statistic $z = 3.06$. P-value = 0.002 to three decimals. Reject H_0 ; there is sufficient evidence that the mean height has changed. Reject for significance levels greater than the P-value.